AI-Based Cloud Resource Monitoring & Predictive Optimization

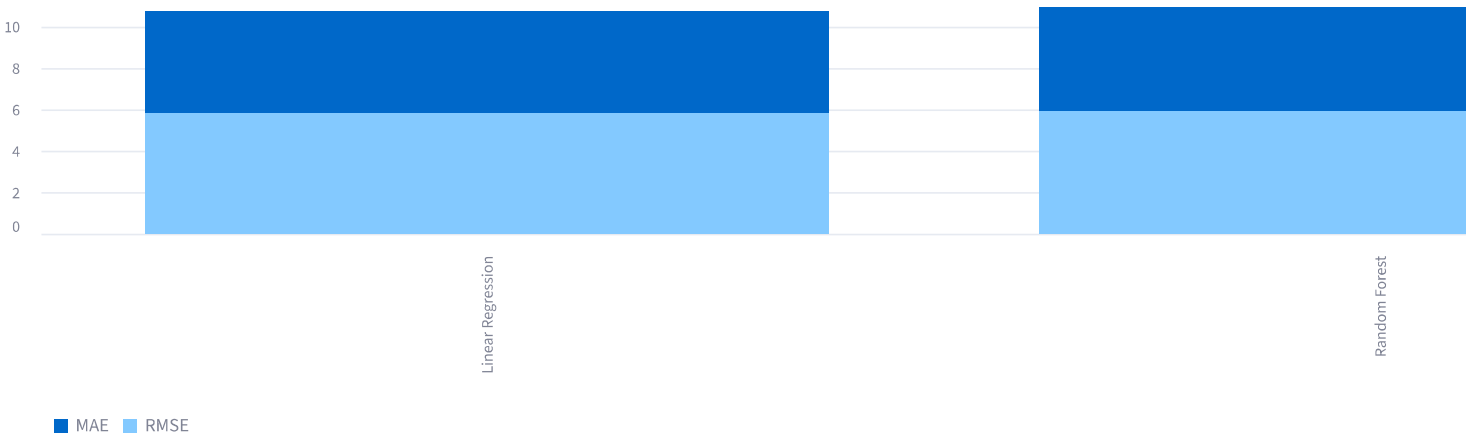
Machine learning for cloud resource usage prediction.

1. Cloud Resource Data

	hour	cpu_usage	memory_usage	request_count
0	0	39.9632	61.7321	972
1	1	86.0571	25.4691	1921
2	2	68.5595	30.5059	1396
3	3	57.8927	78.406	161
4	4	22.4815	59.4179	695
5	5	22.4796	20.5978	828
6	6	14.6467	26.5957	1465
7	7	79.2941	63.1276	496
8	8	58.0892	20.329	798
9	9	66.6458	30.4525	1142

2. Model Performance Comparison

Model	MAE	RMSE	R2 Score
Linear Regression	4.917	5.812	
Random Forest	5.022	5.89	



3. Predict Future CPU Usage

Current CPU Usage (%)

75

Memory Usage (%)

70

Request Count

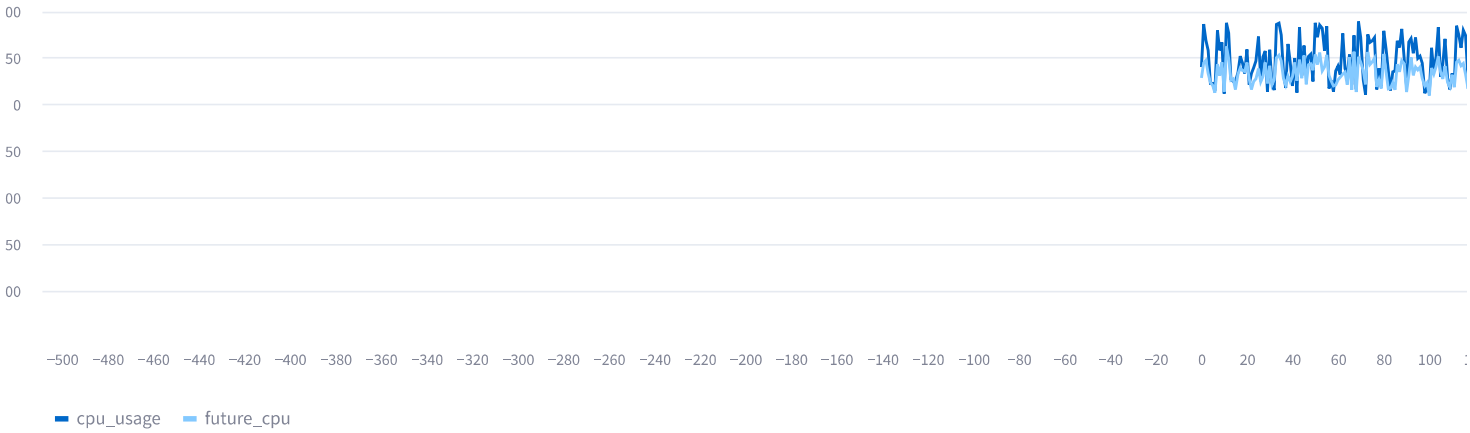
1500

Choose Prediction Model

Linear Regression

Predict CPU Usage

4. CPU Usage Overview



Download Resource Dataset

Educational prototype using synthetic data. It does not monitor live AWS resources.